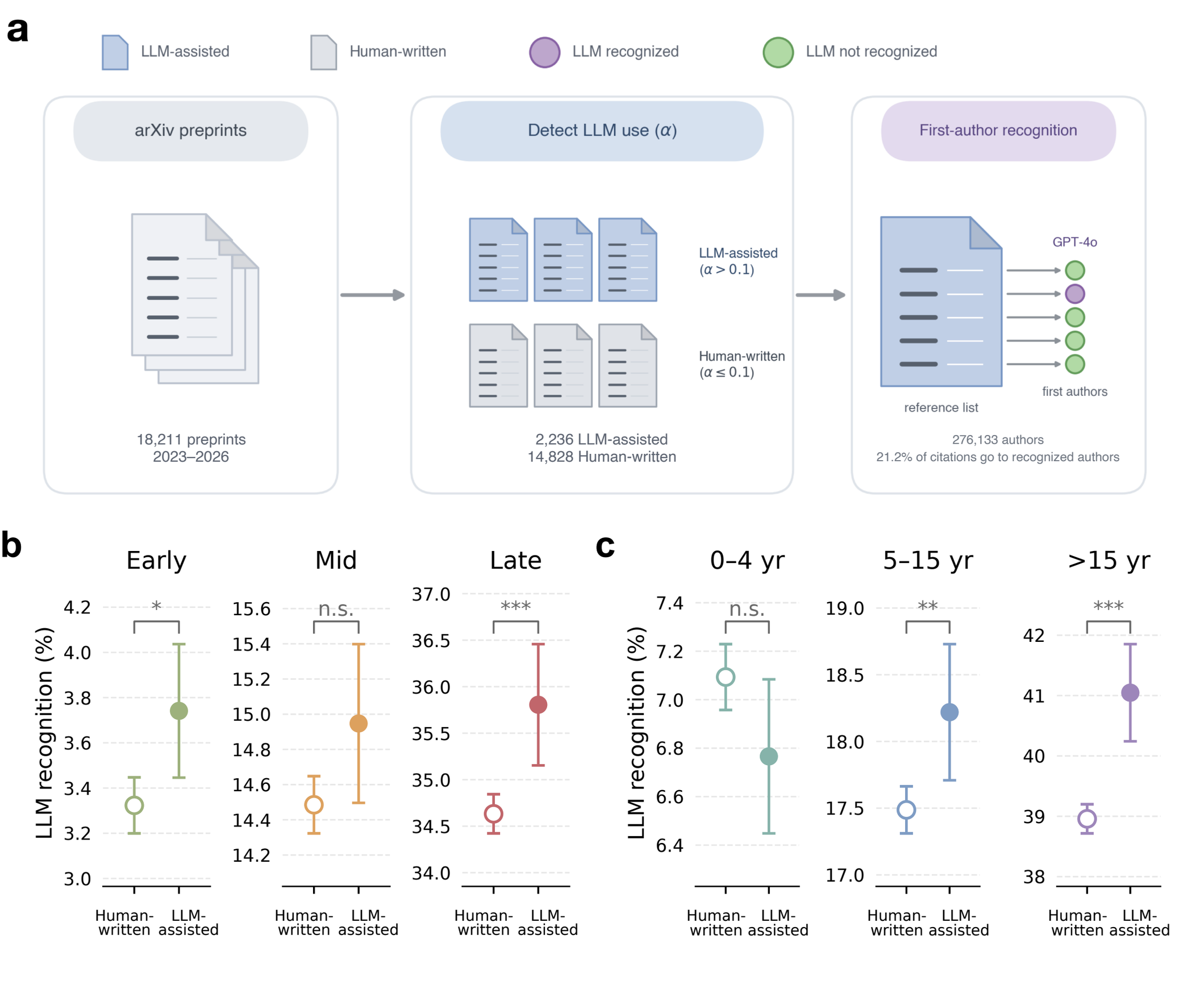

a
LLM-assisted
Human-written
LLM recognized
LLM not recognized
arXiv preprints
18,211 preprints
2023–2026
Detect LLM use (α)
LLM-assisted
(α > 0.1)
Human-written
(α ≤ 0.1)
2,236 LLM-assisted
14,828 Human-written
First-author recognition
GPT-4o
first authors
reference list
276,133 authors
21.2% of citations go to recognized authors
b
Early
Mid
Late
LLM recognition (%)
*
n.s.
***
Human-written
LLM-assisted
c
0–4 yr
5–15 yr
>15 yr
**